THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	$3b + 4 = 19$
2	**(b)** 5	Number of binary digits of the largest weight to be measured
3	**(a)** 32_5	Convert every number to decimal and compare
4	**(b)** 4	$77 = 1001101_2$
5	**(c)** MCMXCIV	Break 1994 into thousands, hundreds, tens and units
6	**(a)** 11000_2	Add 1 and carry in base 2
7	**(b)** 3	The last place has value 3, so the number is a multiple of 3
8	**(a)** 6	$2b + 5 = 17$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$; R is the method used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	XL = 50 − 10 = 40, so A is true; R is also true but does not explain why XL means 40.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2b + 3 = 13$ (1); $b = 5$ (1).	1+1
12	$90 = 2 \times 45 + 0$; $45 = 2 \times 22 + 1$; $22 = 2 \times 11 + 0$; $11 = 2 \times 5 + 1$; $5 = 2 \times 2 + 1$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (1); 1011010_2 has 4 ones (1).	1+1
13	CXXVII = 127 (1); 1111111_2 (1).	1+1
14	$5b + 2 = 37$ (1); $b = 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$45 = 101101_2$ (1); weights used: 1, 4, 8, 32 g (1); each 1 means the weight is used (1).	1+1+1
16	Decimal values 98 and 57 (1); sum 155 (1); 1110_5 (1).	1+1+1
17	Each of the 8 places has 2 choices, so $2^8 = 256$ numbers (2); largest 255 (1).	2+1
	OR: $77 = 1001101_2$ (1); weights used: 1, 4, 8, 64 g (1); each 1 means the weight is used (1).	1+1+1
18	Decimal values 205 and 134 (1); sum 339 (1); 523_8 (1).	1+1+1
	OR: Each of the 5 places has 3 choices, so $3^5 = 243$ numbers (2); largest 242 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $1000 + 900 + 80 + 5 = 1985$ (1). (ii) $10 + 10 + 4 = 24$ (1). (iii) MMXXIV (1). (iv) $2024 - 1985 = 39$ (1); XXXIX (1).	1+1+1+2
20	Digits are all less than b, so $b > 5$ (1); $1b^2 + 4b + 5 = 65$ (2); trying $b = 6$ gives $36 + 24 + 5 = 65$ (1); so $b = 6$ (1).	1+2+1+1
	OR: Decimal values 300 and 149 (1); difference 151 (2); 227_8 (1); check by converting back (1).	1+2+1+1

Mistakes to avoid:

- Repeating V, L or D, or writing IC, IL, XM for 99, 49, 990; use XCIX, XLIX and CMXC.
- Reading the remainders from the first to the last when converting decimal to another base; they must be read from the **last to the first**.
- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4	150 = 10010110_2
2	**(a)** MMCDXLIX	Break 2449 into thousands, hundreds, tens and units
3	**(a)** 11100_2	Add 1 and carry in base 2
4	**(b)** 5	The last place has value 5, so the number is a multiple of 5
5	**(b)** 7	$4b + 3 = 31$
6	**(a)** 6	Number of binary digits of the largest weight to be measured
7	**(a)** 76_8	Convert every number to decimal and compare
8	**(b)** 8	255 = 11111111_2
9	**(c)** A is true but R is false.	The Babylonians used base 60 (true); the Maya used base 20, so R is false.
10	**(d)** A is false but R is true.	99 is written XCIX; IC is not a valid numeral, so A is false, while R is a true rule.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$157 = 2 \times 78 + 1$; $78 = 2 \times 39 + 0$; $39 = 2 \times 19 + 1$; $19 = 2 \times 9 + 1$; $9 = 2 \times 4 + 1$; $4 = 2 \times 2 + 0$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (1); 10011101_2 has 5 ones (1).	1+1
12	CCXLIV = 244 (1); 11110100_2 (1).	1+1
13	$4b + 6 = 38$ (1); $b = 8$ (1).	1+1
14	$203 = 2 \times 101 + 1$; $101 = 2 \times 50 + 1$; $50 = 2 \times 25 + 0$; $25 = 2 \times 12 + 1$; $12 = 2 \times 6 + 0$; $6 = 2 \times 3 + 0$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (1); 11001011_2 has 5 ones (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$102 = 1100110_2$ (1); weights used: 2, 4, 32, 64 g (1); each 1 means the weight is used (1).	1+1+1
16	Decimal values 40 and 29 (1); sum 69 (1); 2120_3 (1).	1+1+1
17	Each of the 4 places has 5 choices, so $5^4 = 625$ numbers (2); largest 624 (1).	2+1
	OR: $89 = 1011001_2$ (1); weights used: 1, 8, 16, 64 g (1); each 1 means the weight is used (1).	1+1+1
18	Decimal values 131 and 85 (1); sum 216 (1); 1000_6 (1).	1+1+1
	OR: Each of the 5 places has 4 choices, so $4^5 = 1024$ numbers (2); largest 1023 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $13 = 1101_2$ (1). (ii) 8, 4 and 1 kg (1). (iii) $27 = 11011_2$, i.e. $16 + 8 + 2 + 1$ (1). (iv) The largest five-digit binary number $11111_2 = 16 + 8 + 4 + 2 + 1 = 31$ (1), and every number from 1 to 31 can be written with at most five binary digits, i.e. as a sum of distinct weights from 1, 2, 4, 8, 16 (1).	1+1+1+2
20	Digits are all less than b, so $b > 7$ (1); $2b^2 + 5b + 7 = 175$ (2); trying $b = 8$ gives $128 + 40 + 7 = 175$ (1); so $b = 8$ (1).	1+2+1+1
	OR: Decimal values 77 and 35 (1); difference 42 (2); 1120_3 (1); check by converting back (1).	1+2+1+1

Mistakes to avoid:

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- Reading the remainders from the first to the last when converting decimal to another base; they must be read from the **last to the first**.
- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	The last place has value 6, so the number is a multiple of 6
2	**(c)** 8	$5b + 6 = 46$
3	**(a)** 8	Number of binary digits of the largest weight to be measured
4	**(a)** 13_5	Convert every number to decimal and compare
5	**(c)** 5	341 = 101010101_2
6	**(a)** MDCXLIX	Break 1649 into thousands, hundreds, tens and units
7	**(a)** 110100_2	Add 1 and carry in base 2
8	**(d)** 8	The last place has value 8, so the number is a multiple of 8
9	**(a)** Both A and R are true and R is the correct explanation of A.	$13 = 8 + 4 + 0 + 1$, so $13 = 1101_2$; the division method in R produces it.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$4 + 2 + 1 = 7$ so A is true; R is true but is not the reason for the value 7.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CLXIX = 169 (1); 10101001_2 (1).	1+1
12	$6b + 1 = 55$ (1); $b = 9$ (1).	1+1
13	$118 = 2 \times 59 + 0$; $59 = 2 \times 29 + 1$; $29 = 2 \times 14 + 1$; $14 = 2 \times 7 + 0$; $7 = 2 \times 3 + 1$; $3 = 2 \times 1 + 1$; $1 = 2 \times 0 + 1$ (1); 1110110_2 has 5 ones (1).	1+1
14	CCCXXX = 330 (1); 101001010_2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$111 = 1101111_2$ (1); weights used: 1, 2, 4, 8, 32, 64 g (1); each 1 means the weight is used (1).	1+1+1
16	Decimal values 70 and 53 (1); sum 123 (1); 1323_4 (1).	1+1+1
17	Each of the 4 places has 6 choices, so $6^4 = 1296$ numbers (2); largest 1295 (1).	2+1
	OR: $58 = 111010_2$ (1); weights used: 2, 8, 16, 32 g (1); each 1 means the weight is used (1).	1+1+1
18	Decimal values 150 and 99 (1); sum 249 (1); 504_7 (1).	1+1+1
	OR: Each of the 3 places has 8 choices, so $8^3 = 512$ numbers (2); largest 511 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 0, 1, 2, 3, 4 (1). (ii) $2 \times 5^3 + 1 \times 5^2 + 4 \times 5 + 3$ (1). (iii) $250 + 25 + 20 + 3 = 298$ (1). (iv) $59 = 2 \times 25 + 1 \times 5 + 4$, so $59 = 214_5$ (1+1).	1+1+1+2
20	Digits are all less than b, so $b > 3$ (1); $3b^2 + 2b + 1 = 57$ (2); trying $b = 4$ gives $48 + 8 + 1 = 57$ (1); so $b = 4$ (1).	1+2+1+1
	OR: Decimal values 150 and 71 (1); difference 79 (2); 1033_4 (1); check by converting back (1).	1+2+1+1

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